On Finite-Size Fluctuations at the 3-Core Transition in Clustered Random Graphs

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***Abstract*—[Abstract text.]**

1. Introduction

Repeatedly removing vertices with fewer than three neighbours can empty a simple graph or leave a substantial part of it intact. Each removal can leave other vertices with too few neighbours and trigger further deletions. The subgraph left when no more vertices can be removed is the 3-core: the largest part of the graph in which every vertex has at least three neighbours within that part. Replacing three by k defines the k-core. In classical random graph models, adding connections can make a substantial 3-core appear abruptly. As the number of vertices grows, the surviving fraction can jump from zero to a positive value at a critical density [1]. This is a discontinuous transition.

The deletion process depends on how neighbours are connected to one another. Triangles link neighbours to each other, creating local clustering. Removing one of these neighbours removes a connection from the other too [2], [3]. An isolated triangle nevertheless has no 3-core, since each vertex has only two neighbours. We therefore study graphs of overlapping triangles, each treated as a community. Each vertex's membership count is drawn from a Poisson distribution [4]. Increasing this distribution's mean adds connections on average.

To compare graphs with similar numbers of connections per vertex, we use a configuration model. It assigns each vertex a number of edge ends, its degree, and joins these ends in random pairs. We choose the degrees so that, as the graphs grow, the proportions of vertices at each degree agree with those in the triangle model. This matches the limiting degree distribution while allowing a different arrangement of connections. The finite graphs need not have identical degrees.

At the same parameter value near the transition, some finite graphs may lose almost all their vertices while others retain a substantial core. An average combines these outcomes, so we measure separately how often a substantial core survives and how its size varies among the graphs that retain one. The second measurement is the conditional core-size distribution. The parameter interval over which survival changes from unlikely to likely is the critical window. Configuration-model theorems describe such finite-size variability under stated assumptions [5].

To relate these two measurements, we start from equations that predict the fraction of vertices left after pruning. For triangles, the equations handle the two linked neighbours together and treat the numbers of remaining neighbours from different communities as independent. Their prediction for the final core remains an approximation. At the predicted threshold, two nonzero solutions of the equations meet at a turning point, called a saddle-node. Just above this point, the predicted surviving fraction increases from its positive threshold value like the square root of the distance from the threshold. We test whether small random shifts of this threshold can explain the variation between graphs. Our main research question is therefore:

> *How do finite-size fluctuations affect the emergence and size of the 3-core in Poisson triangle-community graphs, and to what extent can a stochastic saddle-node approximation predict these fluctuations compared with a degree-matched configuration model?*

For graphs with n vertices, we ask whether the window narrows like $n^{-1/2}$ and the spread of surviving fractions, conditional on a substantial core, decreases like $n^{-1/4}$. The square-root response connects these two proposed scales, which remain hypotheses for triangle graphs. The fraction at the threshold and the coefficient in the square-root relation come from the equations. We estimate the parameters of the random shift using only how often a substantial core survives. Sizes from independent simulations then test the predicted distribution, without fitting its location, amplitude, or spread to those sizes.

The paper focuses on triangle communities and this degree-matched benchmark. After defining the models and their theoretical predictions, we describe the simulations and uncertainty estimates. We then examine survival and conditional size distributions together, and discuss what the approximation explains and which differences remain.

2. Graph models and core observables

2.1. The 3-core and its size

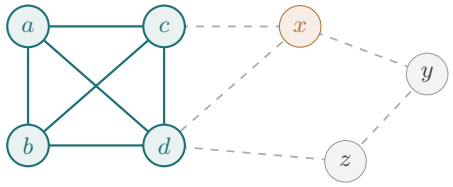
To find the 3-core, remove every vertex with fewer than three remaining neighbours, then repeat: vertices that initially had enough neighbours may lose them during the deletions. A simple undirected graph $G = (V, E)$ has vertex set V and edge set E; edges join distinct vertices and each pair has at most one edge. Write $n = |V| \geq 1$ for its initial number of vertices. For a subset $S \subseteq V$, the induced subgraph $G[S]$ keeps the vertices in S and every edge of G between them. Let $d_{G[S]}(v)$ count the neighbours of v in this subgraph. The simultaneous deletion rounds are

$$S_0 = V, \qquad S_{t+1} = \{v \in S_t : d_{G[S_t]}(v) \geq 3\}, \tag{1}$$

where $t = 0, 1, \ldots$ counts rounds. The sets decrease and eventually stop changing. If S_∞ is the final set, then $C_3(G) = G[S_\infty]$ is the 3-core [1].

This procedure gives the unique largest induced subgraph with minimum degree at least three. Indeed, if $G[W]$ already has that minimum degree, none of its vertices can be the first vertex of W removed: its three neighbours in W would still be present. Thus every such W remains inside the final set, which itself meets the degree condition. This also proves that deleting eligible vertices one at a time, in any order, gives the same answer. The core may be disconnected; we count all its vertices, rather than only its largest connected component.

The implementation uses a queue to avoid rescanning the whole graph after each deletion. Initially queue all vertices of degree below three. Remove the first queued vertex and its incident edges, update the degrees of its remaining neighbours, and queue a neighbour when its degree falls from three to two. Stop when the queue is empty. Each vertex is queued at most once, giving work proportional to $|V| + |E|$. The input graph is left unchanged. Figure 1 illustrates why a single pass would be insufficient. For the configuration multigraph defined in Section 2.3, the same rule uses incident edge ends instead of distinct neighbours, with the degree convention specified there.



Remaining 3-core: $\{a, b, c, d\}$

Round 1: delete y, z. Round 2: delete x.

Figure 1. A fixed illustrative graph, not a random sample. Vertices y, z have degree two and disappear in the first simultaneous round. Vertex x initially has degree three, but loses y and disappears in the second. The four mutually connected vertices a, b, c, d form the final 3-core: its size is four and its fraction of the initial graph is $4/7$.

We use M to distinguish the graph constructions defined below. Their parameter λ controls the mean number of memberships per vertex, as specified in Sections 2.2 and 2.3. Let $G_{n,M}(\lambda)$ be the random input graph and define

$$\begin{aligned} N_{n,M}(\lambda) &= |V(C_3(G_{n,M}(\lambda)))|, \\ R_{n,M}(\lambda) &= N_{n,M}(\lambda)/n. \end{aligned} \tag{2}$$

The denominator includes every initial vertex, including isolates. When n, M, λ are understood, we write simply R. A core is empty if $R = 0$. To separate substantial outcomes from small nonempty cores, choose a fraction $\varepsilon_M \in (0, 1]$ and set

$$\begin{aligned} A_{n,M}(\lambda) &= \{R_{n,M}(\lambda) \geq \varepsilon_M\}, \\ p_{n,M}(\lambda) &= \mathbb{P}(A_{n,M}(\lambda)). \end{aligned} \tag{3}$$

Thus survival means at least $\lceil n\varepsilon_M \rceil$ vertices remain; equality with the cutoff is included. The baseline analyses use $\varepsilon_M = 0.05$ for all three graph conventions, with 0.02 and 0.08 as sensitivity checks in the triangle comparison. These are analysis choices applied after pruning; they do not change the graph or its core. At finite n, $0 < R < \varepsilon_M$ means a nonempty core below the chosen cutoff. A *macroscopic* core instead describes growth with n: its size remains a positive fraction of the input size. A single finite-graph cutoff cannot establish that asymptotic property.

An unconditional mean or distribution includes all outcomes. The conditional distribution keeps only substantial ones:

$$F^+_{n,M}(r; \lambda) = \mathbb{P}(R_{n,M}(\lambda) \leq r \mid A_{n,M}(\lambda)), \tag{4}$$

for $r \in [0, 1]$, provided $p_{n,M}(\lambda) > 0$. Its mean and standard deviation describe sizes among survivors, whereas $p_{n,M}$ describes how often they occur. The unconditional mean also includes below-cutoff cores, which need not be empty. In an empirical sample, estimate $p_{n,M}$ by the proportion meeting the cutoff and $F^+_{n,M}$ by the proportion of those survivors whose fractions are at most r. With no survivors, conditional estimates are unavailable, not zero; a sample standard deviation requires at least two. Existing baseline size summaries are unconditional; conditional distributions are the separate target of the finite-size investigation.

2.2. Poisson triangle communities

A triangle community connects each of three individuals to the other two. Combining such communities allows a vertex to belong to several triangles. It also connects neighbours to one another, introducing the local clustering discussed in the introduction. The construction is a triangle specialization of random intersection graphs with communities [4]. The following specifies the finite simple graph used in the experiments.

Fix $n \geq 1$ individuals, labelled $1, \ldots, n$, and a dimensionless parameter $\lambda \geq 0$. First propose independent membership counts $\widetilde{D}_1, \ldots, \widetilde{D}_n$ with

$$\mathbb{P}(\widetilde{D}_i = r) = e^{-\lambda} \frac{\lambda^r}{r!}, \tag{5}$$

for integers $r \geq 0$. Here λ is the mean number of membership places per individual *before conditioning*. For $\lambda = 0$, every count is zero. If the proposed total is not divisible by three, discard the entire vector and draw again. Denote the accepted counts by $D_1, \ldots, D_n$ and their total by $L = \sum_i D_i$. They have the joint Poisson law conditioned on $L \equiv 0 \pmod 3$. They are therefore not exactly independent Poisson variables at finite n, and their marginal mean need not be exactly λ. No single count is corrected and no leftover memberships are discarded. The implementation aborts with an error after 100 000 unsuccessful proposals; it never substitutes a different construction.

There are now $m = L/3$ communities, each with three labelled places. To assign individuals to them:

1) Make D_i distinct membership tokens labelled i for each individual i. Such a token, often called a membership stub, represents one place still to be assigned.
2) Uniformly shuffle the L tokens. Fill the three places of community one with the first three tokens, community two with the next three, and so on. Thus every matching of labelled tokens to labelled places has the same probability.
3) In each triple, join every pair of distinct individuals, adding an edge only if that pair is not already connected.

If $L = 0$, there are no communities or edges, but all n vertices remain. Uniformity applies to token assignments, not to the resulting simple graphs.

The grouping does not require three different individuals. For example, (i, i, j) creates only the edge $\{i, j\}$, while (i, i, i) creates none. Repeated triples are also allowed. A connection from a vertex to itself is omitted, and every repeated edge is merged into one, both within and between communities. This conversion from membership places to edges is the simple-graph *projection*; the unmerged membership records are retained. Figure 2 shows both overlap and repeated places. Equivalently, one can draw an incidence graph with one type of vertex for individuals and another for communities, joining them once per assigned place. Repeated memberships give repeated incidence edges. Core pruning is performed on the projected graph of individuals, not on this incidence graph, and does not require a prescribed number of whole triangles to survive.

Each membership offers two edge ends, so write $K_i = 2D_i$ for the *nominal* degree, before loops and repeated edges are removed. If κ_i is the actual number of distinct neighbours after projection, then exactly

$$\kappa_i \leq \min\{n - 1, 2D_i\}. \tag{6}$$

Even a vertex with many memberships can have fewer neighbours because some places repeat the same individual. Actual finite degrees can be odd.

For fixed $\lambda > 0$ as $n \to \infty$, the divisibility condition has a vanishing effect on any fixed number of membership counts. Coincidences among the neighbours of a uniformly chosen vertex also become unlikely. Its degree therefore converges in distribution to

$$\begin{aligned} K &= 2D, \qquad D \sim \mathrm{Poisson}(\lambda), \\ \pi_{2r} &:= \mathbb{P}(K = 2r) = e^{-\lambda}\lambda^r/r!, \\ \pi_{2r+1} &= 0, \qquad r = 0, 1, \ldots. \end{aligned} \tag{7}$$

This is the triangle case of the community degree law in [4, Corollary 2.9 and the erasure discussion in Section 2.4]. It gives $\mathbb{E}[K] = 2\lambda$ and $\mathrm{Var}(K) = 4\lambda$. In particular, it is twice a Poisson variable, rather than a Poisson variable of mean 2λ. These are limiting degree statements, not identities for the finite projected degrees or a theorem about the final 3-core.

2.3. The degree-matched configuration model

The triangle construction joins neighbours in prescribed groups. To compare it with connections formed without those groups, give each vertex a number of individual edge ends and randomly join the ends in pairs. This is the configuration model (CM) [1, Section 2]. For the implemented benchmark, independently draw $D_1^{\mathrm{CM}}, \ldots, D_n^{\mathrm{CM}}$ from $\mathrm{Poisson}(\lambda)$ and prescribe degrees $K_i^{\mathrm{CM}} = 2D_i^{\mathrm{CM}}$. These counts are unconditioned: the total degree is automatically even. Thus each prescribed degree has exactly the law in Equation (7), at every n.

Create K_i^{CM} distinct edge ends, or half-edge stubs, at vertex i. Uniformly shuffle all ends and pair consecutive entries. A pair labelled (i, j) becomes an undirected edge, and every end is used. Every matching of the labelled ends is equally likely; the resulting graphs need not be equally likely. A general prescribed sequence with odd total degree is rejected with an error, rather than repaired. That case cannot arise for these twice-Poisson degrees.

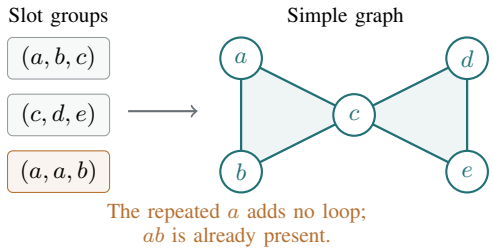


Figure 2. A fixed membership assignment and its simple projection. The triples (a, b, c) and (c, d, e) create two triangles sharing c. The triple (a, a, b) creates no loop and adds no extra copy of ab. Membership counts are $(D_a, D_b, D_c, D_d, D_e) = (3, 2, 2, 1, 1)$, whereas distinct-neighbour degrees are $(2, 2, 4, 2, 2)$. This is an illustration of the construction, not a simulated realization.

Two ends of the same vertex form a *self-loop*. Different pairs can join the same two vertices, forming *parallel edges*. The retained CM graph keeps all of these, so it is a multigraph. Its degree counts incident ends: a loop contributes two and each parallel edge contributes separately. The degree at vertex i is therefore exactly K_i^{CM}. For core pruning, keep a list of neighbours at each vertex, repeating a neighbour once per edge to it and listing the vertex itself twice per loop. The length of this adjacency list is its degree. On removing a vertex, mark it absent first, then reduce each remaining neighbour's degree once per edge to it. Its loops affect no other vertex. Queue a neighbour when its degree crosses from three to two. The definition and order-independence argument in Section 2.1 apply with this degree convention. It need not mean three distinct neighbours: two vertices joined by three parallel edges already form a multigraph 3-core.

From the *same* pairing we also obtain an erased simple CM graph: delete every loop and retain one edge per unordered pair of distinct vertices. Prune this graph separately using distinct-neighbour degrees. Erasing can reduce degrees and core size, and can produce odd degrees. It is different from rejecting an entire pairing until a simple graph is obtained; in particular, its degrees need not equal the prescribed sequence. The retained multigraph is the direct benchmark for the CM theory in Section 3.1. The erased version checks the finite effect of imposing the simple-graph convention used for triangles.

In the existing comparison, triangle and CM graphs use the same n and λ, with independently generated membership vectors and matchings. Only the retained and erased CM outcomes share a pairing. Here *degree-matched* means a common limiting degree distribution, not equal finite degree sequences, empirical histograms, or mean degrees. A planned additional control would use one triangle-conditioned vector D for both constructions and give the CM degrees $2D_i$. It would match nominal degrees exactly, but still not generally match the triangle graph's projected degrees. This shared-vector control is not part of the existing baseline, and its conditioning differs from the unconditioned CM sampler above.

To record finite departures from the common limiting law, let κ_i be the degree of vertex i in the graph being measured: incident ends for the retained CM and distinct neighbours for either simple graph. The broad comparison computes all these degrees, including zeros. It saves their histogram, mean, and population variance:

$$H(j) = \#\{i : \kappa_i = j\}, \qquad j = 0, 1, \ldots,$$
$$\bar{\kappa} = \frac{1}{n}\sum_{i=1}^{n} \kappa_i, \qquad v_\kappa = \frac{1}{n}\sum_{i=1}^{n} (\kappa_i - \bar{\kappa})^2. \tag{8}$$

It also records the fraction $\sum_{j \text{ odd}} H(j)/n$ and the nominal mean $n^{-1}\sum_i K_i$, using the corresponding model's nominal degrees. For either construction, let ℓ count loops in the edge list before erasure and b count non-loop edge copies beyond the first per vertex pair. For triangles, this list contains the three pairs of places from each community. Erasure loses exactly $\ell + b$ edges, reducing total degree by $2(\ell + b)$. These diagnostics quantify projection loss.

Clustering is measured on the simple graph in all cases. If it contains τ triangles and has distinct-neighbour degrees s_i, define

$$\mathcal{T} = \frac{3\tau}{\sum_{i=1}^{n} \binom{s_i}{2}}, \tag{9}$$

with $\mathcal{T} = 0$ when the denominator is zero. The denominator counts pairs of neighbours about a common vertex. Each triangle closes three such pairs, so $\mathcal{T}$ is their connected fraction, called transitivity. Both CM versions share this diagnostic because they share their simple projection; it is measured before core pruning.

This benchmark preserves the limiting number of connections per vertex while removing the requirement that they arrive in triangle communities. The CM can still contain triangles formed by chance. Degree matching does not control every correlation between vertices or every dependence created by communities. Differences between the models therefore reflect those structural changes together, rather than identifying an effect of clustering alone. The next section develops the respective predictions for the measured core fractions, keeping their different theoretical justifications explicit.

3. Critical behavior and finite-size predictions

The models in Section 2 specify how to draw a graph. We now ask what fraction of its vertices can remain after repeated deletion, and how that prediction changes near the transition. There are three levels of justification: established CM core theorems, deductions from the triangle cavity equations, and a proposed finite-size approximation for triangles. Keeping these separate is essential when comparing the predictions with data.

3.1. Configuration-model equations and theorems

An edge can help retain its other endpoint only if the vertex supplying that connection has at least two further surviving connections. Imagine keeping the receiving endpoint present while testing this condition. Let x be the probability that a directed edge supplies such support. This conditional question is called a cavity calculation: the contribution of the receiving edge is excluded while counting the other available connections.

Write K for the limiting CM degree, with probabilities π_k from Equation (7), and let $\mu = \mathbb{E}[K] = 2\lambda$. Following a uniformly chosen edge reaches a degree-k vertex with probability $k\pi_k/\mu$, because it has k possible incident ends. This is the *size-biased* degree law. For fixed $\lambda > 0$, a bounded neighbourhood of the CM becomes a tree as n grows. Its different outward branches are independent in the

limit. Given degree k, the number of other supporting edges is therefore $\mathrm{Bin}(k-1,x)$, a binomial variable with $k-1$ trials and success probability x. Consequently

$$x = F_\lambda(x) := \sum_{k\geq 1} \frac{k\pi_k}{\mu} \mathbb{P}\{\mathrm{Bin}(k-1,x) \geq 2\}. \tag{10}$$

At a uniformly chosen vertex all k incident edges are available, giving

$$\mathcal{R}_{\mathrm{CM}}(x,\lambda) = \sum_{k\geq 0} \pi_k \mathbb{P}\{\mathrm{Bin}(k,x) \geq 3\}. \tag{11}$$

The predicted density is $\rho_{\mathrm{CM}}(\lambda) = \mathcal{R}_{\mathrm{CM}}(x(\lambda),\lambda)$ at the selected solution.

These sums have short closed forms. The function $G_0(t) = \mathbb{E}[t^K] = \exp\{\lambda(t^2-1)\}$ encodes the degree probabilities as its coefficients. The corresponding function after following an edge is $G_1(t) = G_0'(t)/\mu = t\exp\{\lambda(t^2-1)\}$. The probabilities of zero and one successes are $G_1(1-x)$ and $xG_1'(1-x)$. Subtracting them from one gives

$$F_\lambda(x) = 1 - e^{-\lambda x(2-x)}[1 + 2\lambda x(1-x)^2]. \tag{12}$$

For the density, subtract the probabilities of zero, one, and two successes using G_0, xG_0', and $x^2G_0''/2$:

$$\mathcal{R}_{\mathrm{CM}}(x,\lambda) = 1 - e^{-\lambda x(2-x)}\big[1 + 2\lambda x(1-x) + \lambda x^2 + 2\lambda^2 x^2(1-x)^2\big]. \tag{13}$$

The zero solution always exists. To select the solution corresponding to pruning, start with all connections available, $x_0 = 1$, and repeatedly set $x_{t+1} = F_\lambda(x_t)$. Since F_λ is increasing, this sequence decreases to its largest fixed point. Starting at zero would miss a positive core. Above the critical point, the CM law of large numbers identifies Equation (13) at this largest, crossing root as the limiting core fraction; below it the fraction tends to zero [1, Condition 2.2 and Theorem 2.3]. The twice-Poisson degrees satisfy the required degree and moment conditions. That theorem's crossing condition fails at the critical tangency, so it does not assign a deterministic density there. Critical fluctuations require Section 3.4 instead. At $\lambda = 0$ both models are edgeless and their core fractions are zero.

3.2. Joint cavity equations for triangles

The triangle graph contains short cycles even as n grows. Its two neighbours in one community are connected, so treating their edges as independent would change the pruning problem. Instead, retain the incidence representation from Section 2.2. Its local limit is a tree of individual and community vertices [4, Theorem 2.14]. Independence then applies between separate outward community branches.

Following a membership reaches an individual with membership count D^* weighted in proportion to its count. For $D \sim \mathrm{Poisson}(\lambda)$,

$$\mathbb{P}(D^* - 1 = j) = \frac{(j+1)\mathbb{P}(D = j+1)}{\lambda} = e^{-\lambda}\frac{\lambda^j}{j!}. \tag{14}$$

Thus both a root and an individual reached through one community have a Poisson number of available other communities, with mean λ.

Consider a triangle containing v, u_1, u_2, and temporarily keep v present. Let S_i count neighbours offered to u_i through other communities, evaluated with u_i itself kept available. Let $J \in \{0,1,2\}$ count how many of u_1, u_2 can remain jointly and thus supply neighbours to v. If both $S_i \geq 1$, both remain: each has the other two triangle vertices and an external neighbour. If one $S_i = 0$, that individual is deleted, and the other remains exactly when its external count is at least two. All other cases give $J = 0$. This local calculation retains the edge between u_1 and u_2 throughout; it does not demand that an entire triangle survive.

Write $q_j = \mathbb{P}(J = j)$ and $p_j = \mathbb{P}(S = j)$, where S is the external support count on one limiting branch. Independence of the two external branches gives

$$q_2 = (1-p_0)^2, \qquad q_1 = 2p_0(1-p_0-p_1), \qquad q_0 = 1 - q_1 - q_2. \tag{15}$$

A vertex receives a Poisson number of these messages. The counts of messages of sizes one and two are independent Poisson variables with means λq_1 and λq_2. Therefore S has the distribution of $A + 2B$, where $A \sim \mathrm{Poisson}(\lambda q_1)$ and $B \sim \mathrm{Poisson}(\lambda q_2)$ are independent. Enumerating the ways to obtain support zero, one, or two yields

$$p_0 = e^{-\lambda(q_1+q_2)}, \qquad p_1 = \lambda q_1 p_0, \qquad p_2 = \big[\lambda q_2 + \tfrac{1}{2}(\lambda q_1)^2\big]\, p_0. \tag{16}$$

Together, Equations (15) and (16) close the equations in (q_1, q_2). A root survives when its total support is at least three, so

$$\mathcal{R}_\triangle(q,\lambda) = 1 - p_0 - p_1 - p_2, \qquad q = (q_1, q_2). \tag{17}$$

Pruning starts with every triangle offering two neighbours, $(q_1, q_2) = (0, 1)$. Iterate the probability update from that state. The quantities that decrease are the tails $q_1 + q_2 = \mathbb{P}(J \geq 1)$ and $q_2 = \mathbb{P}(J \geq 2)$; q_1 alone can increase when two supports become one. Monotonicity of the local deletion rule makes this iteration select the largest message law in these two tails, and hence the physical density prediction.

The calculation is exact for the stated local rule on the limiting incidence tree. Using it for the finite graph's terminal core additionally assumes that the large-graph limit can be combined with arbitrarily many pruning rounds. Local convergence controls each fixed depth, and does not establish that interchange. We call the resulting $\rho_\triangle(\lambda)$ a cavity prediction. Motif-aware core calculations provide related context [3], but do not supply a fluctuation theorem for this particular ensemble.

3.3. Critical points and the saddle-node expansion

A positive solution first becomes available when two solution branches meet at a turning point, called a saddle-node. For the CM set $\mathcal{G}_{\mathrm{CM}}(x,\lambda) = F_\lambda(x) - x$. Its critical point satisfies

$$F_{\lambda_c}(x_c) = x_c, \qquad \partial_x F_{\lambda_c}(x_c) = 1. \tag{18}$$

For fixed $x \in (0,1)$, $F_\lambda(x)$ increases strictly with λ. Solving $F_\lambda(x) = x$ defines a parameter curve $\Lambda(x)$ whose interior minimum gives Equation (18). Its parameter diverges at both ends. Moreover $F_\lambda(x) = (3\lambda+2\lambda^2)x^2+O(x^3)$ near zero, so a positive root cannot merge continuously with zero at finite λ. The unique critical contact and its nonzero curvature are justified in Section A.1.

For triangles, eliminate the two message probabilities by putting $u = p_0 = e^{-z}$, $z > 0$. At a fixed point,

$$q_2 = (1-u)^2, \qquad q_1 = \frac{2u(1-u)}{1+2\lambda u^2}. \tag{19}$$

The remaining equation is

$$\mathcal{G}_\triangle(z,\lambda) = H(z,\lambda) := \lambda(q_1+q_2) - z = 0. \tag{20}$$

Its derivative $H_\lambda = (1-u)^2 + 2u(1-u)/(1+2\lambda u^2)^2$ is positive. Multiplying by $1+2\lambda u^2$ gives a quadratic in λ, whose positive root is

$$\begin{aligned} \Lambda(z) &= \frac{2z}{B(z) + \sqrt{B(z)^2 + 8u^2(1-u)^2 z}}, \\ B(z) &= 1 - u^2 - 2u^2 z, \qquad u = e^{-z}. \end{aligned} \tag{21}$$

Every positive solution lies on this curve. At fixed λ, larger z means larger values of both q_2 and $q_1 + q_2 = z/\lambda$, so the largest z root gives the pruning-selected law. The criticality equations are $H(z_c,\lambda_c) = H_z(z_c,\lambda_c) = 0$. Since $\Lambda(z) \sim (2z)^{-1/2}$ near zero and $\Lambda(z) \sim z$ at infinity, the first solution occurs at a positive interior minimum. The update Jacobian at zero is $\left(\begin{smallmatrix} 0 & 2\lambda \\ 0 & 0 \end{smallmatrix}\right)$; its eigenvalues are zero, excluding a finite-parameter merger with the zero solution. The predicted density at onset is therefore positive. The observed single-fold shape of the explicit triangle curve is numerically checked; an analytic uniqueness theorem is not claimed.

For either model write $y = x$ or z for its scalar coordinate and $\mathcal{R}_M(y,\lambda)$ for its density map, composing Equation (19) with Equation (17) for triangles. All derivatives below hold the other argument fixed. At the critical point both models have

$$\begin{gathered} \mathcal{G}_M = \mathcal{G}_{M,y} = 0, \\ \mathcal{G}_{M,\lambda} > 0, \quad \mathcal{G}_{M,yy} < 0, \quad \mathcal{R}_{M,y} > 0. \end{gathered} \tag{22}$$

Let $\delta = \lambda - \lambda_c > 0$ and $v = y - y_c$. Taylor expansion gives

$$\begin{aligned} 0 = {} & \mathcal{G}_{M,\lambda}\delta + \tfrac{1}{2}\mathcal{G}_{M,yy}v^2 \\ & + O(\delta^2 + |v|\delta + |v|^3). \end{aligned} \tag{23}$$

TABLE 1. CRITICAL VALUES AND DENSITY COEFFICIENTS COMPUTED FROM THE MODEL EQUATIONS. THE TRIANGLE ROW IS A CAVITY PREDICTION. DIGITS EXPRESS NUMERICAL PRECISION FOR THESE EQUATIONS, NOT FINITE-GRAPH ACCURACY.

Model	λ_c	ρ_c	a
CM multigraph	1.287720	0.137889	0.317959
Triangle cavity	1.228598	0.081112	0.278553

The larger branch consequently has $v = b_M\sqrt{\delta} + O(\delta)$, where $b_M = \sqrt{-2\mathcal{G}_{M,\lambda}/\mathcal{G}_{M,yy}}$. Expanding the density in turn gives the required result:

$$\begin{aligned} \rho_M(\lambda) &= \rho_{c,M} + a_M\sqrt{\lambda - \lambda_{c,M}} + O(\lambda - \lambda_{c,M}), \\ a_M &= \mathcal{R}_{M,y}\sqrt{-2\mathcal{G}_{M,\lambda}/\mathcal{G}_{M,yy}}. \end{aligned} \tag{24}$$

Here $\rho_{c,M} > 0$ is the positive branch's right-hand limit, distinct from the zero solution below the threshold. The direct dependence of $\mathcal{R}_M$ on λ contributes only at order δ. Thus a coefficient derived for a message coordinate must be multiplied by $\mathcal{R}_{M,y}$ before predicting a core fraction.

The numerical procedure first brackets the minimum of Λ and then bisects its derivative, rather than reporting a grid point as the threshold. For CM, implicit differentiation gives $\Lambda' = (1 - \partial_x F_\lambda)/(\partial_\lambda F_\lambda)$; for triangles it gives $\Lambda' = -H_z/H_\lambda$. Above the fold, bisection on the larger-coordinate side selects the positive branch. Residuals in the original equations check the result. Table 1 records the evaluated constants. Analytic derivative formulas are given in Section A.1. Independent centred differences and direct branch solutions verify them: $[\rho_M(\lambda_c+\delta) - \rho_c]/\sqrt{\delta}$ approaches a_M, while the unscaled remainder is of order δ. Figure 3 shows both the jump and this local limit. The near-double root magnifies roundoff at extremely small offsets, so the checks use a sequence of resolvable offsets and derivative step sizes (Section A.2).

3.4. Critical windows and fluctuation scales

Finite graphs with identical n and λ need not produce the same core. Their sampled memberships differ, and even fixed memberships can be matched differently. Near a turning point, a small perturbation can either leave a positive branch available or remove it altogether. This motivates treating the distance from criticality as a random quantity.

The CM supplies an established benchmark for this reasoning. The appropriate results are the prescribed-degree critical theorem and the random-degree conditioning argument of [5, Theorem 3.5 and Section 8]. Its Erdős–Rényi-specific constants cannot be used for twice-Poisson degrees. Here is their specialization to our retained CM, with the checks and variance calculation detailed in Section A.1.

Let $\lambda_n = \lambda_c + u/\sqrt{n}$ for a fixed real offset u. For a prescribed degree k, put $h_k(x) = \mathbb{E}[Y\mathbf{1}_{\{Y\geq 3\}}]$, where $Y \sim \mathrm{Bin}(k,x)$ and $\mathbf{1}$ is the indicator of the stated event. In the theorem's auxiliary pruning process, the expected

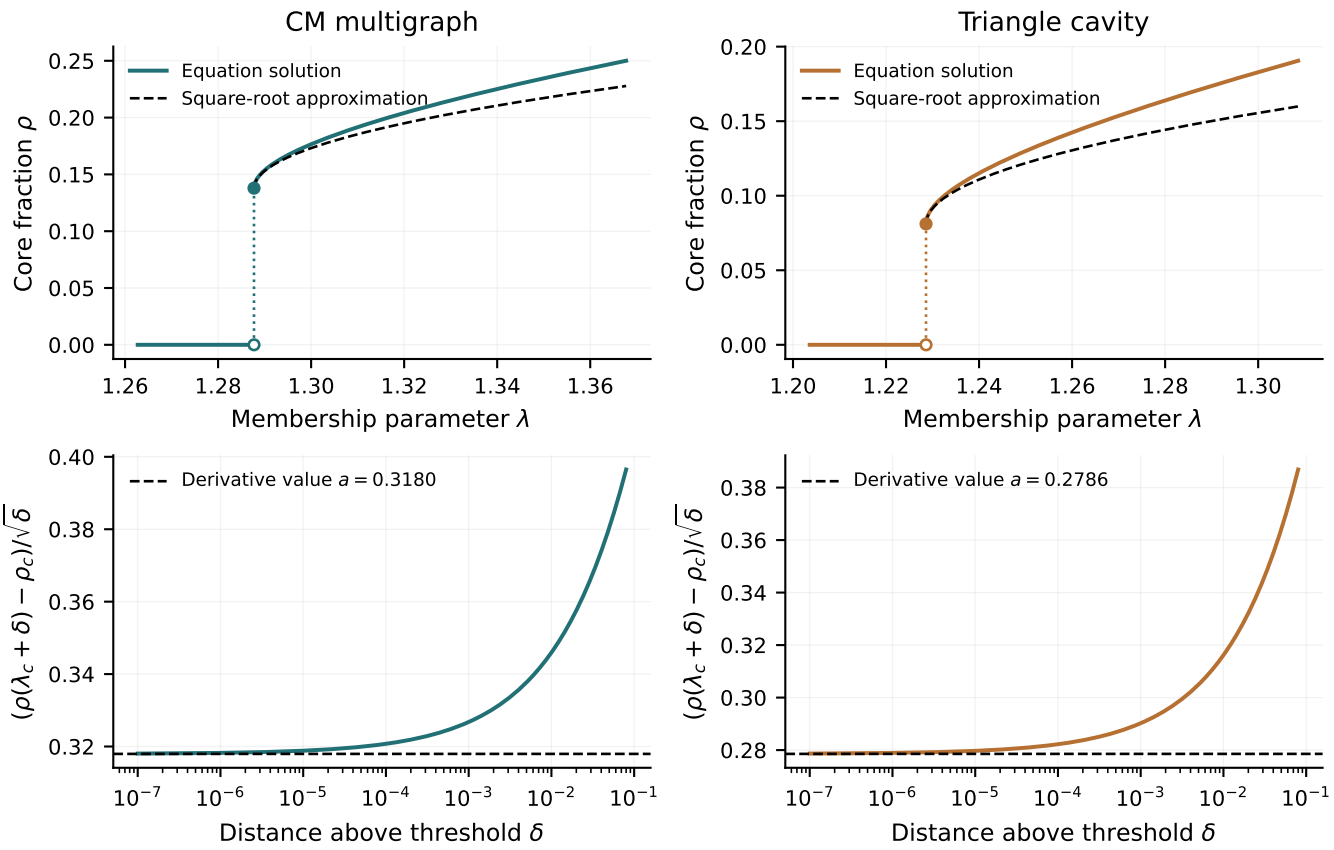


Figure 3. Deterministic theory near the transition. Upper panels compare the maximal-branch density with its leading square-root approximation. Dotted vertical segments indicate jumps from the separate zero branch; filled markers denote the positive right-hand limits. Lower panels divide the density increase by $\sqrt{\delta}$, where $\delta = \lambda - \lambda_c > 0$. The ratios approach the derivative-based coefficients (dashed lines), while higher-order terms become visible farther above onset. No graph simulations or fitted density parameters enter this figure.

number of remaining edge ends per input vertex is the mean degree times x^2. The quantity $h_k(x)$ counts ends at vertices with at least three remaining ends. Their difference counts ends at vertices still eligible for deletion. Running out of such ends can stop the process with a positive core. The corresponding empirical difference is

$$\ell_n(x) = \frac{1}{n}\sum_{i=1}^{n}[K_i x^2 - h_{K_i}(x)]. \tag{25}$$

Its population version is $\ell(x,\lambda) = 2\lambda x[x - F_\lambda(x)]$. At (x_c,λ_c) it has a quadratic minimum. Define $c = -\partial_\lambda \ell(x_c,\lambda_c) > 0$ and $\beta = \partial_x^2 \ell(x_c,\lambda_c) > 0$. The first measures the change in the minimum's height with λ; the second measures its curvature.

Sampling the degrees makes the minimum fluctuate: if $\bar{x}_n$ is its nearby minimizer, then

$$\sqrt{n}\,\ell_n(\bar{x}_n) \xrightarrow{d} -cu + U, \qquad U \sim N(0, v_{\text{degree}}). \tag{26}$$

The pruning theorem adds an independent centred Gaussian contribution from pairing, with variance v_{pair}. For our degree law the two evaluated variances are approximately 0.249099 and 0.107879, respectively. Set

$$\tau_{\text{CM}} = \frac{\sqrt{v_{\text{degree}} + v_{\text{pair}}}}{c} \approx 1.95625. \tag{27}$$

Let $Z \sim N(0,1)$ and $W = u + \tau_{\text{CM}} Z$. Writing $Y_{n,\text{CM}} = n^{1/4}(R_{n,\text{CM}} - \rho_{c,\text{CM}})$, the critical-window limits become

$$\begin{aligned} &\mathbb{P}(A_{n,\text{CM}}(\lambda_n)) \longrightarrow \Phi(u/\tau_{\text{CM}}), \\ &Y_{n,\text{CM}} \mid A_{n,\text{CM}} \xrightarrow{d} a_{\text{CM}}\sqrt{W} \mid \{W > 0\}. \end{aligned} \tag{28}$$

Here Φ is the standard normal distribution function and $\xrightarrow{d}$ denotes convergence in distribution. Any fixed cutoff $0 < \varepsilon_{\text{CM}} < \rho_{c,\text{CM}}$ selects the same asymptotic substantial-core event. Both sources of randomness matter: using pairing noise alone would give a different limit for the existing experiments, which resample degrees on every trial.

For triangles, suppose the scalar equation has a random perturbation of order $n^{-1/2}$ near its fold. Writing its leading noise as $n^{-1/2}\xi$ in Equation (23) replaces δ by $\delta + n^{-1/2}\xi/\mathcal{G}_{M,\lambda}$. If the scaled perturbation has a nondegenerate limiting distribution, a change in λ of order $n^{-1/2}$ competes with it. The fold then turns this into a coordi-

nate and density displacement of order $n^{-1/4}$ on surviving outcomes. Thus we test a window of width $O(n^{-1/2})$ and

$$R_{n,M} - \rho_{c,M} = O_p(n^{-1/4}) \text{ conditional on } A_{n,M}. \quad (29)$$

The notation means that $n^{1/4}(R_{n,M} - \rho_{c,M})$ stays bounded in probability under the conditional laws. Multiplying by n gives core-count fluctuations of order $n^{3/4}$ about $n\rho_{c,M}$. These statements apply within the critical window, not at every parameter. Unconditionally, the distribution also contains outcomes near zero, far from ρ_c. Neither the noise order nor Gaussianity follows from the deterministic square root. They are additional triangle hypotheses; local incidence convergence proves neither. The CM constants also do not automatically apply to its erased projection.

3.5. A stochastic prediction for surviving core sizes

To obtain a prediction that can be tested without fitting the sizes it aims to explain, let survival observations determine an effective threshold's centre and spread. For each n, M, let $\mu_{n,M}$ be its centre in membership parameter units and $w_{n,M} > 0$ its width. Propose the occurrence curve

$$\mathbb{P}(A_{n,M}(\lambda)) \approx \Phi\left(\frac{\lambda - \mu_{n,M}}{w_{n,M}}\right). \quad (30)$$

This is equivalent to a normal random effective threshold. At a given λ, abbreviate $m = \sqrt{n}(\lambda - \mu_{n,M})$ and $s = \sqrt{n}\, w_{n,M}$, and define

$$V = m + sZ, \qquad Z \sim N(0,1). \quad (31)$$

The approximation identifies survival with $V > 0$. Its conditional density prediction is

$$Y := n^{1/4}(R_{n,M} - \rho_{c,M}) \mid A_{n,M} \;\approx\; a_M\sqrt{V} \mid \{V > 0\}. \quad (32)$$

The same random distance explains both whether a core appears and how much its fraction exceeds the jump. Theory supplies $\rho_{c,M}$ and a_M; only $\mu_{n,M}, w_{n,M}$ are estimated from survival indicators. For the CM asymptotic benchmark, $\mu_{n,\mathrm{CM}} = \lambda_c$ and $\sqrt{n}\, w_{n,\mathrm{CM}} = \tau_{\mathrm{CM}}$ recover Equation (28). Allowing a finite-size centre and width makes possible corrections measurable; their fitted values can be compared with that limit. The width hypothesis is that $\sqrt{n}\, w_{n,M}$ approaches a positive constant. Using λ_c as the centre of the same scaled window additionally requires $\sqrt{n}(\mu_{n,M} - \lambda_{c,M})$ to remain bounded. This centre-drift condition is a separate assumption to be checked.

The model gives a full distribution, not just a mean. For $y \geq 0$, its conditional distribution function is

$$F_Y(y) = 1 - \frac{\Phi\big((m - (y/a_M)^2)/s\big)}{\Phi(m/s)}, \quad (33)$$

and $F_Y(y) = 0$ for $y < 0$. For a probability $q \in (0,1)$ its quantile is

$$Q_Y(q) = a_M\sqrt{m - s\Phi^{-1}\big((1-q)\Phi(m/s)\big)}. \quad (34)$$

Means and variances follow by integrating this law; explicit moment formulas are in Section A.1. Convert back to a core fraction by $R \approx \rho_{c,M} + n^{-1/4}Y$. The result is generally non-Gaussian because it takes a square root after truncating the random distance at zero.

After the survival curve is calibrated, the conditional prediction has no further fit parameters. Independent validation graphs then test its conditional means, standard deviations, quantiles, and entire distribution. Complete membership vectors, including repeated matchings or paired model outcomes from each vector, must remain within one calibration or validation group. Calibration uncertainty must be propagated into the predictions; conditional sizes must not be used to adjust them. These statistical procedures belong to the next experimental phase and are not implemented by the present deterministic calculations.

The approximation assumes that one effective threshold displacement captures the leading size variability, that its normal occurrence model is adequate, and that the local fold remains the relevant response. The cutoff must lie below the positive jump. Finite graphs can have substantial core fractions below ρ_c, whereas the leading law in Equation (32) has $Y \geq 0$; persistent discrepancies there, or in spread and tails, can challenge the approximation. Such failure is a possible result of the experiment, not a reason to refit the conditional distribution to agree with it.

4. Numerical methodology

4.1. Sampling, pruning, and numerical checks

[Graph algorithms, solvers, and reproducibility checks.]

4.2. Critical-window design and controls

[Scaled grids, paired controls, and pilot-based sample planning.]

4.3. Survival calibration and validation

[Indicator-only fitting, data splits, and prediction uncertainty.]

4.4. Distributions, uncertainty, and sensitivity

[Windows, full conditional distributions, and robust uncertainty.]

5. Finite-size results and model comparisons

5.1. Limiting densities and numerical checks

[Existing density benchmarks, critical values, and solver checks.]

5.2. Survival probabilities and window scaling

[Occurrence curves, centre drift, and evidence on window widths.]

5.3. Conditional sizes and predictive accuracy

[Conditional size distributions and held-out prediction errors.]

5.4. Degree controls and sources of fluctuations

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6. Discussion

6.1. Interpreting finite-size fluctuations

[Scaling evidence, approximation accuracy, and unexplained spread.]

6.2. Limitations and possible extensions

[Finite-size and theoretical limitations, then scoped future work.]

7. Conclusion

[Answers to the research questions and main takeaways.]

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Appendix A. Supplementary derivations and checks

A.1. Derivatives and fluctuation constants

For the CM, write $F(x,\lambda) = F_\lambda(x)$, $r = 1-x$, and $E = \exp\{-\lambda x(2-x)\}$. Differentiating the closed forms gives

$$F_x = 2\lambda x r E(3+2\lambda r^2), \tag{35}$$

$$F_\lambda = x^2 E[3-2x+2\lambda(2-x)r^2], \tag{36}$$

$$F_{xx} = 2\lambda E\big[(1-2x)(3+2\lambda r^2) - 2\lambda x r^2(5+2\lambda r^2)\big]. \tag{37}$$

The binomial-tail derivative also gives $\mathcal{R}_{\mathrm{CM},x} = \lambda x F_x$. At the fold this is simply $\lambda_c x_c > 0$. These formulas establish the coefficient in Equation (24) without fitting a branch curve.

The uniqueness condition needed by the CM critical theorem can be checked analytically here. For $0 < \lambda \le 1.3$,

$$\begin{aligned}(\log F_x)_{xx} &= -x^{-2} - r^{-2} + 2\lambda + \frac{4\lambda(3-2\lambda r^2)}{(3+2\lambda r^2)^2} \\ &\le -8 + \tfrac{10}{3}\lambda < 0.\end{aligned} \tag{38}$$

Thus F_x has a single maximum and vanishes at both endpoints. The residual $F - x$ consequently has at most one interior local maximum. For $\lambda \le 1/2$, the bound $F_x \le \lambda(3+2\lambda)/2 \le 1$ excludes positive roots. At $x = 0.44, \lambda = 1.3$, $F > x$: equivalently $e^{0.89232} > 1.3587584/0.56$, already certified by the first five terms of the exponential series. Hence first contact occurs for $1/2 < \lambda_c < 1.3$, at the unique interior maximum, with $F_{xx} < 0$. The residual is nonpositive elsewhere at criticality. Therefore $\ell(x,\lambda_c) = 2\lambda_c x(x-F)$ is nonnegative, has exactly one positive zero, and has positive second derivative there. This verifies the global minimum hypothesis, rather than inferring it from a plot.

For triangles use $u = e^{-z}$, $d = 1+2\lambda u^2$, and $C = 2u - 1 + 2\lambda u^2$ in this derivative calculation. With q_1, q_2 from Equation (19),

$$\begin{aligned} q_{1,z} &= 2uC/d^2, \qquad q_{2,z} = 2u(1-u), \\ q_{1,zz} &= -\frac{2u(4u-1+6\lambda u^2)}{d^2} + \frac{16\lambda u^3 C}{d^3}, \\ q_{2,zz} &= 2u(2u-1), \\ H_{zz} &= \lambda(q_{1,zz} + q_{2,zz}). \end{aligned} \tag{39}$$

For the density put $Q = \lambda(q_1 + q_2)$ and $T = \lambda q_1$. The smooth extension away from the solution curve is $\mathcal{R}_\triangle = 1 - e^{-Q}(1 + Q + T^2/2)$, giving

$$\mathcal{R}_{\triangle,z} = e^{-Q}[Q_z(Q + T^2/2) - TT_z]. \tag{40}$$

Retaining e^{-Q} when differentiating keeps this the original density map; $e^{-Q} = u$ is an identity only on $H = 0$. At the critical solution, $H_\lambda \approx 0.346630$, $H_{zz} \approx -0.702172$, and $\mathcal{R}_{\triangle,z} \approx 0.280338$, all nonzero with the signs required by Equation (22). These numerical checks verify a local nondegenerate fold of the stated equations. They do not prove a terminal-core limit for finite triangle graphs.

For the random-degree CM specialization, if $\lambda_n \to \lambda_c > 0$ then $K_i = 2\,\mathrm{Poisson}(\lambda_n)$ has limiting probabilities Equation (7), even total degree, and $\mathbb{E}[A^{K_i}] = \exp\{\lambda_n(A^2 - 1)\} < \infty$ for every fixed $A > 1$. There is positive probability of degree greater than three. The empirical degree proportions converge in probability, and Markov's inequality gives $\sum_i A^{K_i} = O_p(n)$. These verify the random-degree moment assumptions in [5, Condition 8.1]. The unique nondegenerate minimum was verified above. The paper's Theorem 3.5 concerns a configuration multigraph with prescribed degrees. Section 8 formally states its random-degree extension for a graph uniform over simple graphs with the given degrees. Its conditioning argument applies directly to the prescribed-degree multigraph theorem as well; this is the extension used here. Uniform-simple conditioning is different from the erased control in our experiments.

For completeness, the constants can be computed with elementary binomial moments. In the following expectations $K \sim 2\,\mathrm{Poisson}(\lambda_c)$, $x = x_c$, and $Y \mid K \sim \mathrm{Bin}(K, x)$. Let $g(Y) = Y\mathbf{1}_{\{Y \geq 3\}}$ and $h_k = \mathbb{E}[g(Y) \mid K = k]$. Then

$$\begin{aligned} v_{\mathrm{degree}} &= \mathrm{Var}(Kx^2 - h_K), \\ v_{\mathrm{pair}} &= \mathbb{E}[\mathrm{Var}(g(Y) \mid K)] - 2\mu x^2(1 - x^2), \\ c &= \mu x \partial_\lambda F(x, \lambda_c), \qquad \mu = 2\lambda_c. \end{aligned} \tag{41}$$

The second identity specializes the paper's pairing variance [5, Eq. (5.47)]. Its unrestricted total-end variance is $\sigma_{WW} = 2\mu x^2(1 - x^2)$, and its heavy-end variance is $\sigma_{HH} = \mathbb{E}[\mathrm{Var}(g(Y) \mid K)]$. Composing binomial thinning and weighting the tail counts by $3, 1, 1, \ldots$ reduces its Eq. (5.40) to

$$\sigma_{WH} = 2x^3 h'(x) \int_x^1 t^{-3}\, dt = x(1 - x^2)h'(x), \tag{42}$$

where $h(x) = \mathbb{E}[h_K(x)]$. At the fold, $h'(x) = 2\mu x$, so $\sigma_{WH} = \sigma_{WW}$ and $\sigma_{LL} = \sigma_{WW} - 2\sigma_{WH} + \sigma_{HH}$ reduces to Equation (41). In that paper the symbol for mean graph degree is our μ, not our membership parameter λ.

For $k \geq 3$, put $P_1 = kx(1 - x)^{k-1}$ and $P_2 = \binom{k}{2}x^2(1 - x)^{k-2}$. Then

$$\begin{aligned} h_k &= kx - P_1 - 2P_2, \\ \mathbb{E}[g(Y)^2 \mid K = k] &= kx(1 - x) + k^2x^2 - P_1 - 4P_2. \end{aligned} \tag{43}$$

Both moments are zero for $k < 3$. Summing these expressions over Poisson memberships computes the variances without simulating either degrees or pairings. The resulting $c \approx 0.305420$ and total variance 0.356978 give Equation (27).

The elementary central limit theorem for independent degree contributions in Equation (25) gives Equation (26) at x_c. Its derivative there is $O_p(n^{-1/2})$, while its second derivative tends to $\beta > 0$. Thus $\bar{x}_n - x_c = O_p(n^{-1/2})$ and replacing the empirical minimum by its value at x_c changes its height by only $O_p(n^{-1})$. The empirical density at that minimizer differs from ρ_c by $O_p(n^{-1/2})$, negligible on the $n^{-1/4}$ density scale. Conditional on the degrees, the pruning theorem supplies independent pairing noise. The normal contributions combine, and its density factor is $\mathcal{R}_{\mathrm{CM},x}\sqrt{2c/\beta} = a_{\mathrm{CM}}$, yielding Equation (28). A fixed cutoff below ρ_c is asymptotically equivalent to nonemptiness because surviving fractions concentrate at ρ_c in this window.

The planned shared-vector CM additionally conditions the total $L = \sum_i D_i$ on $L \equiv 0 \pmod 3$. This changes finite distributions, but not these leading Gaussian constants. To see this, set $\omega = e^{2\pi i/3}$ and use $\mathbf{1}_{\{L \equiv 0\}} = (1 + \omega^L + \omega^{2L})/3$. For any finite vector $f(D)$ of the degree contributions needed above, the characteristic function of its centred sum has, after conditioning, three terms. The ordinary term gives its central limit theorem. In each remaining term, the single-observation factor has modulus tending to $|\mathbb{E}[\omega^{jD}]| = e^{-3\lambda_c/2} < 1$, $j = 1, 2$; its nth power vanishes. The conditioning probability is

$$\begin{aligned} \mathbb{P}(L \equiv 0 \pmod 3) = \tfrac{1}{3}[1 + 2e^{-3n\lambda_n/2} \\ \times \cos(\sqrt{3}n\lambda_n/2)] \longrightarrow \tfrac{1}{3}. \end{aligned} \tag{44}$$

Hence the same Gaussian degree limits hold, and the moment bounds survive conditioning on this event of probability bounded away from zero. Pairings remain uniform conditional on each vector. Repeated pairings of one fixed vector share its degree contribution U; their within-vector variation therefore contains only pairing noise. They are not independent samples from the full ensemble law in Equation (28).

Finally, the stochastic law in Equation (32) has, for $r = 1, 2$,

$$\mathbb{E}[Y^r] = \frac{a_M^r}{s\Phi(m/s)} \int_0^\infty v^{r/2}\phi((v - m)/s)\, dv, \tag{45}$$

where ϕ is the standard normal density. Its variance is $\mathbb{E}[Y^2] - \mathbb{E}[Y]^2$, and the second moment simplifies to $a_M^2[m + s\phi(m/s)/\Phi(m/s)]$. Thus the predicted conditional spread is fixed by theory and the occurrence fit; it has no independent fit parameter.

A.2. Independent numerical checks

Notebooks 05 and 08 retain the shared equation solvers and add analytic partial derivatives and density coefficients.

The deterministic comparison notebook evaluates them and generates the table, figure, and diagnostic CSVs. The bisection stopping tolerances are 10^{-15} for CM and 2×10^{-15} for triangles, relative to the coordinate scale, with finite iteration limits and errors on failure. These are root-location tolerances, not error bounds on finite-graph predictions.

For CM, centred differences use coordinate steps from 10^{-2} to 10^{-4}; for triangles they use 10^{-3} to 10^{-5}. They independently differentiate the residual and density, with additional second differences of the triangle residual itself. Agreement improves until subtraction roundoff becomes relevant. Checks at noncritical points also verify that the partial derivatives hold the correct argument fixed. Direct solutions at offsets down to 10^{-8} give density ratios 0.317987 (CM) and 0.278590 (triangles), approaching the respective coefficients in Table 1. The remainder divided by δ approaches approximately 0.2772 and 0.3653 over the resolved range, consistent with the stated $O(\delta)$ remainder.

The CM variance calculation agrees using truncations at 40 and 60 Poisson memberships. Direct binomial probability sums independently check the moment identities in Equation (43). An additional 48-, 96-, and 192-point Gaussian quadrature evaluation of the paper's unsimplified covariance integral agrees with $\sigma_{WH} = 0.8014616885$ to within 6×10^{-15}. These are deterministic numerical cross-checks, not random-graph measurements. Fresh-kernel tests also retain the existing local triangle-rule, probability-sum, branch-selection, and pruning checks.

Appendix B. Experiment settings and sensitivity checks

B.1. Configurations and data provenance

[Complete experimental settings and reproduction information.]

B.2. Additional diagnostics and sensitivity

[Distributional checks, uncertainty, and cutoff sensitivity.]